THE DERIVED PULLBACK OF AN M-PSEUDO-COHERENT OBJECT ALONG A MORPHISM OF RINGED SITES IS M-PSEUDO-COHERENT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof is the author's complete argument: the reduction along coverings of both sites, the localisation of the strictly perfect witness, the computation with the distinguished triangle and the vanishing of the pulled-back cone, and the final use of the strictly-perfect pullback lemma are all written out.

Editorial note. The label lemma-pseudo-coherent-pullback names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The statement and proof are copied from the pinned chapter file; the six cited inputs are kept as external pointers with their canonical Stacks tags.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- m-pseudo-coherent objects of D(O) on a ringed site. Same chapter, the definition of m-pseudo-coherence by the existence of a strictly perfect complex mapping to the object with cohomology isomorphisms above m and a surjection in degree m. Supplies both the hypothesis on E and the property verified for Lf^*E.
- Local character of pseudo-coherence. Same chapter, the independence of the representative and the locality of m-pseudo-coherence under restriction along a covering. Supplies the reduction from an arbitrary U to the members of a covering and to the localisations C/U, D/V.
- Strictly perfect complexes are pulled back to strictly perfect complexes. Same chapter, the pullback lemma for strictly perfect complexes along a morphism of ringed sites. Supplies the final step: Lf^* of a strictly perfect complex is strictly perfect, so it witnesses pseudo-coherence of Lf^*E.
- Vanishing of the left-derived pullback in high degrees. Derived Categories, the vanishing statement for the left derived functor applied to a complex with vanishing cohomology above a bound. Supplies the vanishing H^j(Lf^*C) = 0 for j >= m, which is what makes the pulled-back morphism a witness.

Source reference. Original lemma 05TC: sources/stacks-chow/tags.txt, source lines 7476–7476. Prerequisite pointer invoked at proof line(s) 12167; it names a standard

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fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 08CC: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 12066–12066. Prerequisite pointer invoked at proof line(s) 12124; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 08H3: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 11755–11755. Prerequisite pointer invoked at proof line(s) 12173; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 08H2: sources/stacks-chow/tags.txt, source lines 10765–10765. Prerequisite pointer invoked at proof line(s) 12121; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 04J0: sources/stacks-chow/tags.txt, source lines 5921–5921. Prerequisite pointer invoked at proof line(s) 12133; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 04IY: sources/stacks-chow/tags.txt, source lines 5919–5919. Prerequisite pointer invoked at proof line(s) 12130; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let $(f, f^\sharp) : (\mathcal{C}, \mathcal{O}_\mathcal{C}) \to (\mathcal{D}, \mathcal{O}_\mathcal{D})$ be a morphism of ringed sites. Let E be an object of $D(\mathcal{O}_\mathcal{C})$. If E is m-pseudo-coherent, then Lf^*E is m-pseudo-coherent.*

Proof. Say f is given by the functor $u : \mathcal{D} \to \mathcal{C}$. Let U be an object of $\mathcal{C}$. By Sites, Lemma 08H2 we can find a covering $\{U_i \to U\}$ and for each i a morphism $U_i \to u(V_i)$ for some object V_i of $\mathcal{D}$. By Lemma 08CC it suffices to show that $Lf^*E|_{U_i}$ is m-pseudo-coherent. To do this it is enough to show that $Lf^*E|_{u(V_i)}$ is m-pseudo-coherent, since $Lf^*E|_{U_i}$ is the restriction of $Lf^*E|_{u(V_i)}$ to $\mathcal{C}/U_i$ (via Modules on Sites, Lemma 04IY). By the commutative diagram of Modules on Sites, Lemma 04J0 it suffices to prove the lemma for the morphism of ringed sites $(\mathcal{C}/u(V_i), \mathcal{O}_{u(V_i)}) \to (\mathcal{D}/V_i, \mathcal{O}_{V_i})$. Thus we may assume $\mathcal{D}$ has a final object Y such that $X = u(Y)$ is a final object of $\mathcal{C}$.

Let $\{V_i \to Y\}$ be a covering such that for each i there exists a strictly perfect complex $\mathcal{F}_i^\bullet$ of $\mathcal{O}_{V_i}$-modules and a morphism $\alpha_i : \mathcal{F}_i^\bullet \to E|_{V_i}$ of $D(\mathcal{O}_{V_i})$ such that $H^j(\alpha_i)$ is an isomorphism for $j > m$ and $H^m(\alpha_i)$ is surjective. Arguing as above it suffices to prove the result for $(\mathcal{C}/u(V_i), \mathcal{O}_{u(V_i)}) \to (\mathcal{D}/V_i, \mathcal{O}_{V_i})$. Hence we may assume that there exists a strictly perfect complex $\mathcal{F}^\bullet$ of $\mathcal{O}_\mathcal{D}$-modules and a morphism $\alpha : \mathcal{F}^\bullet \to E$ of $D(\mathcal{O}_\mathcal{D})$ such that $H^j(\alpha)$ is an isomorphism for $j > m$ and $H^m(\alpha)$ is surjective. In this case, choose a distinguished triangle

$$\mathcal{F}^\bullet \to E \to C \to \mathcal{F}^\bullet[1]$$

The assumption on α means exactly that the cohomology sheaves $H^j(C)$ are zero for all $j \geq m$. Applying Lf^* we obtain the distinguished triangle

$$Lf^*\mathcal{F}^\bullet \to Lf^*E \to Lf^*C \to Lf^*\mathcal{F}^\bullet[1]$$

By the construction of Lf^* as a left derived functor we see that $H^j(Lf^*C) = 0$ for $j \geq m$ (by the dual of Derived Categories, Lemma 05TC). Hence $H^j(Lf^*\alpha)$ is

an isomorphism for $j > m$ and $H^m(Lf^*\alpha)$ is surjective. On the other hand, since $\mathcal{F}^\bullet$ is a bounded above complex of flat $\mathcal{O}_\mathcal{D}$-modules we see that $Lf^*\mathcal{F}^\bullet = f^*\mathcal{F}^\bullet$. Applying Lemma 08H3 we conclude. □